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in Further Mathematics (9FM0)

Paper 02 Core Pure Mathematics 2

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General comments

This paper allowed learners to demonstrate their knowledge and apply it across a range of specification topics. Some questions included structured parts, enabling learners to attempt later parts even where earlier parts had not been completed. In some responses, learners used results from previous “show that” questions effectively to support later work. Learners should be reminded that, in “show that” questions, sufficient working must be shown because the answer is given. These points were particularly relevant to Questions 4, 6 and 8.

Question 1

This question allowed learners who were secure with matrix multiplication and inverses to demonstrate their understanding across all parts. Many learners made progress throughout the question, with algebraic accuracy being important in supporting successful responses.

In part (i)(a), most learners correctly evaluated the matrix product $\mathbf{P}^2$. In stronger responses, the matrix multiplication was carried out accurately and simplified to give a scalar multiple of the identity matrix. Common errors included sign slips involving $-\sqrt{3}$ terms, particularly when calculating the off-diagonal entries.

Part (i)(b) was answered less successfully. Learners who had correctly obtained $\mathbf{P}^2$ generally recognised the transformation as an enlargement. In some responses identified only one aspect of the transformation, for example, did not mention the centre of enlargement. Some learners described the transformation as two stretches, rather than as the single transformation required by the question.

In part (ii)(a), most learners knew the process to find the inverse of a *2 x 2* matrix and correctly found the determinant, as $k-4$. A common error was to write the determinant as $4-k$, with a few consequences such as sign errors in the final answer, switching and negating the wrong diagonals, negating all the elements, multiplying by the determinant, or omitting the determinant.

Part (ii)(b) proved to be the most demanding part of the question. Stronger responses substituted their expression for into the given matrix equation and compared corresponding entries systematically. These learners usually recognised that the off-diagonal entries provided the simplest route to a solution. Algebraic manipulation errors were common, particularly when clearing fractions. Some learners who obtained a quadratic equation in k did not reject the value that did not satisfy the original matrix equation.

Question 2

This question assessed learners' understanding of the method of differences and their ability to apply a general result to a related finite sum. Responses that included a clear partial fraction decomposition and a structured argument were generally successful.

In part (a), many learners recognised that the first step was to split the expression into partial fractions. In stronger responses, learners obtained the correct expression and wrote out several consecutive terms of the series to show the cancellation explicitly. These responses usually gained full credit.

A common error was an incorrect partial fraction decomposition, often arising from algebraic slips when equating coefficients. Some responses reached the correct decomposition but did not show the method of differences clearly enough, instead giving the result with little or no justification. Learners are reminded that, when instructed to use the method of differences, they should include sufficient intermediate working to show the cancellation process.

Some learners correctly obtained an expression involving n but they were not successful when simplifying the remaining terms to the required form. Algebraic slips were seen when combining fractions leading to incorrect values of a and b.

In part (b), learners who obtained a correct answer to part (a) usually went on to answer correctly, showing the required evidence. Some learners applied the correct technique with an incorrect final answer to part (a) and were only able to gain the method mark. Stronger responses split into the required sums $\sum_{r=1}^{30} - \sum_{r=1}^{9}$, substituted appropriately into the result from part (a) and obtained the correct answer. Some responses gave the correct value using a calculator only, without the required supporting work, and were not awarded marks here.

Question 3

This question allowed learners to demonstrate their understanding of hyperbolic functions, including their representation in terms of exponential functions and the use of hyperbolic identities.

In part (a), most learners were able to use a definition of $\tanh x$, in terms of exponentials, to form an expression in e^x or e^{2x}. Responses using the e^x form was more successful where a common factor was identified and cancelled. From there, learners either recognised the exponential form of $\sinh x$ and $\cosh x$ or expanded the brackets to achieve $\frac{1}{2}\left(e^{2x} - e^{-2x}\right)$ leading to $\sinh 2x$, the required

answer. Responses using the e^{2x} form often required more careful algebraic manipulation to reach $\frac{e^{4x}-1}{2e^{2x}}$ which is also a form of $\sinh 2x$.
Stronger responses used a structured approach, simplifying the numerator and denominator separately before combining them into a single fraction and reaching the correct answer. Some responses expanded the brackets, which made it more difficult to identify and cancel common factors.

Many responses did not reach a form equivalent to the required result. A small number of learners used hyperbolic identities rather than the exponential definition required by the question, so did not gain marks in this part.

In part (b), many learners used the identity in part (a) to establish the main result i.e. $2\cosh^2 2x = -5\sinh 2x$. Many then applied the identity $\cosh^2 2x = 1+\sinh^2 2x$ to form a three-term quadratic in $\sinh 2x$. Most learners then solved the quadratic and went on to use the logarithmic form of expression for $\operatorname{arcsinh} x$ to reach correct solution. Some responses returned to the exponential definitions rather than using the formula book, which often led to more complex algebra and subsequent errors. A common error after solving the quadratic had $\sinh x = -2, -\frac{1}{2}$ instead of $\sinh 2x = -2, -\frac{1}{2}$ which prevented access to the final marks.

The most common difficulty was not using the ‘hence ‘instruction in the question, which limited the progress in this part.

Question 4

In part (a), the differentiation of $\ln(\sec x)$ was generally carried out well. Most learners worked confidently with $\sec x$ and $\tan x$ with only a small number of responses converting to $\sin x$ and $\cos x$. Some responses did not reach the final answer. The most common error occurred when moving from the third differential to the fourth differential, particularly where an incorrect coefficient was introduced (usually with the $\sec^2 2x \tan^2 x$ term). These errors had an impact on later parts of the question. Stronger responses used the product rule clearly, which supported a successful derivation of the required answer.

In part (b), most responses demonstrated a good understanding of the Maclaurin series expansion formula, correctly calculating the values of the derivatives up to the fourth derivative. Some responses included slips in the calculation of the fourth derivative. In some cases, coefficient slips from part (a) did not prevent learners from reaching the required result, as the incorrect term became zero when evaluated; however, these responses did not gain the final accuracy mark.

In part (c), most learners understood the requirement to form the expansion of $\ln(1+2x)$ using the given expansion for $\ln(1+x)$ formula, although some responses used differentiation instead, and did not gain the B mark. Learners are reminded to read the demand of the question carefully. Many learners recognised the requirement for log law application and were able to subtract their two expansions appropriately, although some responses did not apply the log law correctly. A number of responses achieved the correct final expansion, either through fully correct working across the earlier parts or despite a small slip in part (a). In stronger responses, the substitution into the given expansion was shown before simplification. In stronger responses the substitution of $2x$ into the expansion for $\ln(1+x)$ was seen before simplification.
For questions of this type, learners are advised to show the substitution of values into their derivatives, and then into the series expansion.

Question 5

This question assessed learners' ability to integrate expressions leading to inverse trigonometric and hyperbolic functions, and to apply a given hyperbolic substitution accurately. Many responses were successful, particularly in part (ii), with several fully correct solutions seen. Some responses did not include constants of integration, although this was not penalised in this question.

In part (i), most learners gave an answer of the correct form $A \arcsin\left[\mathrm{f}(x)\right]$ and gained the method mark. A common error was including an extra constant factor multiplying an otherwise correct answer. Many forms of the correct answer were seen and scored full marks as simplification was not required. By not factoring out the $\sqrt{25}$ made the algebraic manipulation more challenging. Sometimes candidates did not follow their calculations with the correct argument inside the function or wrote $\operatorname{arsinh}(\mathrm{f}(x))$ or $\operatorname{arcosh}(\mathrm{f}(x))$. Some learners used integration by substitution for this part which was long-winded but generally resulted in the correct answer.

Part (ii) proved more demanding. Stronger responses differentiated the required substitution and substituted in correct to produce an integral in terms of u only. They then use the identity $\cosh^2 u - 1 = \sinh^2 u$ to enable cancellation to take dx place. On a few occasions the square root in the denominator was omitted, and so did not simplify to the required form. Learners who successfully cancelled were able to get the correct answer.

In some responses, learners substituted directly into the fraction without first differentiating the substitution. Although the correct identity was then used to simplify, an additional factor remained in the working. Some responses reached the correct answer from incorrect work.

Overall, learners who demonstrated familiarity with standard integral forms and the basic identities of hyperbolic functions were generally able to gain full credit. Clear algebraic manipulation and careful substitutions back into the original variable were key features of successful responses.

Question 6

Learners were able to access many of the marks in this question and showed a good understanding of the principles of mathematical induction.

Most learners gained credit for a correct evaluation of the base case and then made the appropriate assumption.

Stronger responses showed the use of the inductive hypothesis to write an expression for $\mathbf{M}^{k+1} = \mathbf{M} \times \mathbf{M}^k$ or $\mathbf{M}^k \times \mathbf{M}$ and went on to use the method for matrix multiplication. In other responses there was sufficient method for matrix multiplication to gain the later marks, just writing down the simplified result (we need to see full detail for a proof). Some numerical or notational slips were seen, and again candidates need to work with full accuracy in proof questions.

The final accuracy mark, which required a properly worded conclusion, was challenging to be awarded. A common incorrect conclusion was along the lines of "true for $n=1$, $n=k$ and $n=k+1$, therefore true for all n". Some responses simply restated what they had done, e.g. "I have shown it's true for $n=1$, I have assumed it's true for $n=k$, I have shown its true for $n=k+1$. In stronger responses there was a convincing link between "**if true** for $n=k$ **then** true for $n=k+1$".

Question 7

This question was highly accessible, and many learners provided concise solutions that gained full credit across all parts of the question.

In part (a)(i), many learners understood how to find the required vector equation of line. Some responses did not gain this mark because the answer was not written as an equation, missing $r =$ or using λ which was the parameter for the given equation, so they needed to use a different letter. Where vectors were incorrect, the error was most often in the *z* component of the direction vector, which was sometimes given as zero.

In part (a)(ii), most learners used the scalar product with the correct vectors to find the angle accurately. Some arithmetic slips were seen, including multiplying by zero incorrectly. A small number of responses did not give their answer to the nearest degree, as required by the question. Learners are reminded to check the accuracy required in the question.

Some responses used sine rather than cosine, while others found the correct acute angle but then gave the complementary angle as the final answer. Responses that included a sketch were often more successful in this part. A small number of learners used an approach involving the vector product.

Part (b) was a good discriminator. In stronger responses, learners equated the coordinates of the two vector equations and solved consistently, obtaining $\mu = 4$ and $\lambda = 3$. They then verified all three coordinates and gave the common point $(9,10,3)$. Some responses checked only two coordinates, which was not sufficient to show that the paths met, or did not draw an adequate conclusion.

Question 8

This question assessed learners' ability to solve a first-order linear differential equation and find a particular solution. The question was structured in parts to support learners through the process.

In part (a), this part of the question was generally tackled well. When identifying the terms leading to the integrating factor, some responses omitted the negative sign. Marks were more commonly lost where there was insufficient working to prove the given statement. Some responses did not show any intermediate step, $e^{\ln(\sin 2x)^{-1}}$ or $e^{\ln(\operatorname{cosec} 2x)}$ or $(\sin 2x)^{-1}$, to demonstrate the logarithm power law and fully show the connection to $\operatorname{cosec} 2x$.

In part (b), learners who found part (a) challenging often found it more difficult to use the given integrating factor effectively.

Most responses started this part well, although some did not multiply the right-hand side by $\operatorname{cosec} 2x$. Learners typically multiplied through by the given integrating factor from (a) to produce $y \operatorname{cosec} 2x = \int \operatorname{cosec} 2x \tan^2 2x \, dx$. Some responses worked entirely in terms of sine and cosine. While these approaches were valid, they often led to errors when integrating. Stronger responses simplified the integral to $\sec 2x \tan 2x$ and were most successful when integrating the function to the correct form; integration by parts was rarely seen. In responses that were making progress, the constant was usually placed correctly, and learners were then able to rearrange successfully to achieve $y = \ldots$

In part (c), most learners recognised the required approach and were successful using their general solution from part (b). They substituted in the given values for x and y to find the value for the constant of integration. Some responses mixed up the solving steps, achieving $c = -1$ as a common error.

Where an incorrect expression had been obtained, some learners substituted the values and continued, leading to increasingly complex constants and values. Other responses stopped at this stage, while some continued and were able to gain method marks for substituting into their answer from part (b). Many learners were well positioned at this point and obtained the required particular solution.

Question 9

This question centred on a design modelled by an equation in polar coordinates. It assessed learners' ability to find areas and identify when a tangent was parallel to the initial line.

In part (a), many learners understood the requirements of the question and generally recalled the area formula accurately, $\text{area} = \frac{1}{2}\int r^2 dx$, though some responses showed uncertainty about the appropriate limits to use. Using limits of 0 and π instead of $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ with double the area. Most candidates were able to identify $\cos 2\theta = 1 - 2\sin^2\theta$ to replace $\sin^2 x$ and obtained an integrable form. The integration was sound apart from a sign slip when integrating $\sin\theta$ or forgetting to divide by 2 when integrating the $\cos 2\theta$ term. The resulting terms needed careful checking as many seemed to have the correct answer, but the accuracy marks were forfeited after incorrect work. The final mark was correct solution only, so any errors led to this mark not being awarded.

In part (b), the question required the use of calculus, so responses that did not include differentiation did not gain marks. Most learners were able to form an equation for y using $y = r\sin\theta$ and proceed to differentiate. A small number of responses incorrectly used $x = r\cos\theta$, which meant that marks could not be awarded for this part of the question. Learners who differentiated correctly generally continued successfully and gained full credit. A common error seen was not identifying from the diagram that the obtuse angle $\theta = \frac{5\pi}{6}$ was required angle due to *B* being in the second quadrant. This often led incorrectly to $\theta = \frac{\pi}{6}$ being given as the angle. Candidates should be encouraged to use the diagram that is given to them in the question to help with determining angles. Using an angle that was found learners then correctly found the value for a, however this was incorporated into the accuracy with the correct angle. A few responses showed an incorrect answer such as $\left(\frac{5\pi}{6}, a\right)$instead of $\left(a, \frac{5\pi}{6}\right)$. If they had previously written $\theta = \frac{5\pi}{6}$ and $r = a$ the accuracy mark was still awarded.

In part (c), this proved to be a challenging part of the question, including responses that have been successful in earlier parts. Some learners found it difficult to adapt their approach to determine the values of r which corresponded to either point B or D. Responses that obtained a value of r for D were able generally to progress and complete the proof, this part was left unattempted by some learners.

In part (d), as in part (c), this part was left unattempted by some learners. Most learners who attempted this part gained both marks. A common error was finding the unshaded area rather than the shaded area that was required. Some responses did not account for the width of the rectangle and treated the outer shape as a square with $\frac{9a}{2}$ as the side lengths. Some responses also showed errors when calculating a percentage of an amount.